

1) a) For $0 < r < R$:

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} \rightarrow E \cancel{4\pi r^2} = \frac{1}{\epsilon_0} \int \rho(r') d\tau' = \frac{1}{\epsilon_0} \int_0^r \rho(r') 4\pi r'^2 dr'$$

Gauss's Law

For $r \geq R$:

$$Q_{tot} = Q_{enc}(R)$$

$$= 4\pi \epsilon_0 \left(\frac{R^3}{3} - \frac{1}{R} R^4 \right)$$

$$= 4\pi \epsilon_0 \left(\frac{R^3}{3} - \frac{R^3}{4} \right) = \frac{R^3}{12} 4\pi \epsilon_0 = \frac{\pi \epsilon_0 R^3}{3}$$

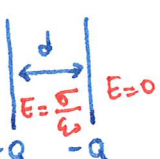
$$Q_{enc}(r) = \frac{\epsilon_0 4\pi}{\epsilon_0} \left\{ \int_0^r r'^2 dr' - \int_0^r \frac{r'^3}{R} dr' \right\}$$

$$= \frac{\epsilon_0 4\pi}{\epsilon_0} \left(\frac{r^3}{3} - \frac{1}{R} \frac{r^4}{4} \right) \Rightarrow E(r) = \frac{\epsilon_0}{\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$$

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} \rightarrow E \cancel{4\pi r^2} = \frac{\pi \epsilon_0 R^3}{3 \epsilon_0} \rightarrow E(r) = \frac{\epsilon_0 R^3}{12 \epsilon_0} \frac{1}{r^2}$$

$$b) E_{in}(R) = \frac{\epsilon_0}{\epsilon_0} \left(\frac{R}{3} - \frac{R^2}{4R} \right) = \frac{\epsilon_0 R}{12 \epsilon_0}$$

$$E_{out}(R) = \frac{\epsilon_0 R^3}{12 \epsilon_0} \frac{1}{R^2} = \frac{\epsilon_0 R}{12 \epsilon_0}$$

2) a)  $E' = \frac{\sigma}{2\epsilon_0}$: Electric field created by a surface charge distribution.

$E = 2 \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$: Electric field between the plates.

$$E = \frac{Q}{A \epsilon_0} \rightarrow E^2 = \frac{Q^2}{A^2 \epsilon_0^2}$$

$$U = \frac{\epsilon_0}{2} \int_{all\ space} E^2 d\tau = \frac{\epsilon_0}{2} \left(\int_{outside} E^2 d\tau + \int_{inside} E^2 d\tau \right) = \frac{\epsilon_0}{2} \int_{inside} \frac{Q^2}{A^2 \epsilon_0^2} d\tau = \frac{Q^2}{2A^2 \epsilon_0} \int_{inside} d\tau$$

volume between the plates

$$= \frac{Q^2}{2A^2 \epsilon_0} A d = \frac{Q^2 d}{2A \epsilon_0} = U$$

b) If the distance is increased from d to $d + \Delta x$ the new energy stored will be

$$U' = \frac{Q^2 (d + \Delta x)}{2A \epsilon_0} \Rightarrow \Delta U = U' - U = \frac{Q^2 (d + \Delta x)}{2A \epsilon_0} - \frac{Q^2 d}{2A \epsilon_0} = \frac{Q^2 \Delta x}{2A \epsilon_0} = \Delta U = W$$

• Since the system is isolated, the work done in increasing the separation is equal to the increase in electrostatic energy.

$$3) a) d\vec{E} = -\frac{1}{4\pi\epsilon_0} \frac{dq}{(d+x)^2} \hat{x} = -\frac{1}{4\pi\epsilon_0} \frac{\lambda(x) dx}{(d+x)^2} \hat{x} = \boxed{-\frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{L} \frac{x}{(d+x)^2} dx \hat{x} = d\vec{E}}$$

$$b) \vec{E} = -\frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{L} \left[\int_0^L \frac{x}{(d+x)^2} dx \right] \hat{x}$$

It is already given.

$$\ln\left(\frac{d+L}{d}\right) + \frac{d}{d+L} - 1$$

$$= -\frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{L} \left[\ln\left(\frac{d+L}{d}\right) + \frac{d}{d+L} - 1 \right] \hat{x}$$

4) Arrangement (a)

$$Q = 3q - q = 2q : \text{monopole moment}$$

$$\vec{p} = 3qa\hat{z} + 0 = 3qa\hat{z} : \text{dipole moment}$$

$$V(r) \cong \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{r} + \frac{3qa \cos\theta}{r^2} \right)$$

Arrangement (b)

$$Q = 3q - q = 2q : \text{monopole moment}$$

$$\vec{p} = 0 - qa(-\hat{z}) = qa\hat{z} : \text{dipole moment.}$$

$$V(r) \cong \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{r} + \frac{qa \cos\theta}{r^2} \right)$$